THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A straight line parallel to the time axis on a velocity-time graph indicates: 1

(a) rest only **(b)** increasing acceleration
(c) retardation **(d)** uniform velocity

2. A ball is thrown vertically upward. At the highest point its velocity is: 1

(a) 10 m/s **(b)** equal to initial **(c)** zero **(d)** maximum

3. The SI unit of velocity is: 1

(a) m/s **(b)** km **(c)** m/s squared **(d)** m

4. A straight line sloping upward on a velocity-time graph indicates: 1

(a) retardation **(b)** uniform acceleration
(c) rest **(d)** uniform velocity

5. In uniform circular motion, the speed is constant but the velocity changes because: 1

(a) the mass changes **(b)** the direction keeps changing
(c) the magnitude changes **(d)** there is no force

6. The SI unit of acceleration is: 1

(a) kg **(b)** m/s squared **(c)** m/s **(d)** m

7. Convert 90 km/h into m/s: 1

(a) 250 **(b)** 90 **(c)** 25 **(d)** 9

8. In uniform circular motion, the velocity at any point is directed: 1

(a) along the tangent **(b)** towards the centre
(c) along the radius **(d)** away from the centre

9. A boy walks 3 m east and then 4 m north. The distance covered is: 1

(a) 12 m **(b)** 7 m **(c)** 5 m **(d)** 1 m

10. A car goes from 10 m/s to 20 m/s in 5 s. Its acceleration is: 1

(a) 0.5 m/s squared **(b)** 10 m/s squared **(c)** 2 m/s squared **(d)** 5 m/s squared

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What does the slope of a velocity-time graph represent? 2

12. Write the three equations of motion for uniform acceleration. 2

13. Can the magnitude of displacement be greater than the distance travelled? Give a reason. 2

14. What does the area under a velocity-time graph represent? How is it found for constant velocity? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A car moving at 20 m/s is brought to rest with a retardation of 5 m/s squared. Find the time taken and the distance covered. 3

16. A bicycle slows from 8 m/s to 4 m/s in 2 s. Find the acceleration and the stopping time from 4 m/s at the same rate. 3

17. State three differences between scalar and vector quantities with examples. 3

OR

A man drives 200 km north in 3 h and then 200 km south in 2 h. Find the average speed and the average velocity. 3

18. Explain uniform circular motion with examples. Why is it an accelerated motion? 3

OR

An object moves 5 m east, then 12 m north. Find the displacement and its direction relative to east. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain distance, displacement, speed, velocity and acceleration with a suitable example of a runner on a circular track. 5

OR

Describe an activity to find the average speed of a student between two points using a stopwatch and measuring tape; discuss sources of error. 5

20. A body is thrown vertically upward with a speed of 40 m/s (g = 10 m/s squared). Find the maximum height, the time to reach it and the total time of flight. 5

OR

A stone is dropped from the top of a tower and takes 4 s to reach the ground. Find the height of the tower, the velocity on reaching the ground and the distance in the last second (g = 10 m/s squared). 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND